

Conformal Transformation of $f(R) = R + \alpha R^2$ to Einstein Frame

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1. Action in the Jordan Frame

We begin with the action for $f(R)$ gravity in the Jordan frame:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} f(R) = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} (R + \alpha R^2) \quad (1)$$

where:

- $\kappa^2 = 8\pi G$
- g is the determinant of the metric in the Jordan frame

This action includes higher derivatives due to the R^2 term, indicating a second-order modified gravity theory.

2. Rewriting as a Scalar-Tensor Theory

We introduce an auxiliary scalar field χ and rewrite the action as:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} [f'(\chi)(R - \chi) + f(\chi)] \quad (2)$$

For $f(R) = R + \alpha R^2$, we compute:

$$f'(\chi) = 1 + 2\alpha\chi \quad (3)$$

$$f(\chi) = \chi + \alpha\chi^2 \quad (4)$$

Substituting into the action:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} [(1 + 2\alpha\chi)(R - \chi) + \chi + \alpha\chi^2] \quad (5)$$

$$= \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} [(1 + 2\alpha\chi)R - \alpha\chi^2] \quad (6)$$

Now define the scalar field $\phi \equiv f'(\chi) = 1 + 2\alpha\chi$, so that:

$$\chi = \frac{\phi - 1}{2\alpha} \quad (7)$$

Substitute this into the action:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} \left[\phi R - \frac{(\phi - 1)^2}{4\alpha} \right] \quad (8)$$

This is a scalar-tensor form of the action, where ϕ is non-minimally coupled to gravity.

3. Conformal Transformation to Einstein Frame

We now perform a conformal transformation:

$$g_{\mu\nu} \rightarrow \tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}, \quad \text{with } \Omega^2 = \phi \quad (9)$$

Under this transformation:

- $\sqrt{-g} \rightarrow \sqrt{-\tilde{g}} \Omega^{-4}$
- $R \rightarrow \tilde{R} - 6\tilde{g}^{\mu\nu} \tilde{\nabla}_\mu \tilde{\nabla}_\nu \ln \Omega - 6\tilde{g}^{\mu\nu} (\tilde{\nabla}_\mu \ln \Omega)(\tilde{\nabla}_\nu \ln \Omega)$

Using this, the Ricci scalar term becomes:

$$\phi R \rightarrow \phi \left[\frac{1}{\phi} \left(\tilde{R} - 6\frac{1}{\phi} \tilde{\nabla}^\mu \phi \tilde{\nabla}_\mu \phi + \dots \right) \right] \quad (10)$$

Neglecting the total derivative term (a boundary term), and simplifying, we obtain:

$$S = \int d^4x \sqrt{-\tilde{g}} \left[\frac{1}{2\kappa^2} \tilde{R} - \frac{3}{4\kappa^2 \phi^2} \tilde{g}^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2\kappa^2} \cdot \frac{1}{\phi^2} \cdot \frac{(\phi - 1)^2}{4\alpha} \right] \quad (11)$$

4. Canonical Normalization of Scalar Field

To get a canonical kinetic term, we define a new scalar field φ such that:

$$\frac{d\varphi}{d\phi} = \sqrt{\frac{3}{2}} \cdot \frac{1}{\kappa} \cdot \frac{1}{\phi} \quad \Rightarrow \quad \varphi = \sqrt{\frac{3}{2}} \cdot \frac{1}{\kappa} \ln \phi \quad (12)$$

Thus,

$$\phi = e^{\sqrt{\frac{2}{3}} \kappa \varphi} \quad (13)$$

Now rewrite the action in terms of φ :

$$S = \int d^4x \sqrt{-\tilde{g}} \left[\frac{1}{2\kappa^2} \tilde{R} - \frac{1}{2} \tilde{g}^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - V(\varphi) \right] \quad (14)$$

where the potential is:

$$V(\varphi) = \frac{1}{8\alpha\kappa^2} \left(1 - e^{-\sqrt{\frac{2}{3}} \kappa \varphi} \right)^2 \quad (15)$$

5. Final Result: Action in the Einstein Frame

$$S = \int d^4x \sqrt{-\tilde{g}} \left[\frac{1}{2\kappa^2} \tilde{R} - \frac{1}{2} \tilde{g}^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{8\alpha\kappa^2} \left(1 - e^{-\sqrt{\frac{2}{3}}\kappa\varphi} \right)^2 \right] \quad (16)$$

This is Einstein gravity plus a canonical scalar field φ with a potential. The scalaron φ represents the extra degree of freedom from the R^2 term in the original theory. This potential is well-known from Starobinsky inflation.

Justification for Taking a Constant Scalar Field in the Einstein Frame

In the Einstein frame, the action for the $f(R) = R + \alpha R^2$ model becomes:

$$S = \int d^4x \sqrt{-\tilde{g}} \left[\frac{1}{2} \tilde{R} - \frac{1}{2} \tilde{g}^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - V(\varphi) \right], \quad (17)$$

where φ is the scalar field obtained through a conformal transformation from the original Jordan frame.

Before introducing perturbations, we must specify the *background* configuration $(\tilde{g}_{\mu\nu}^{(0)}, \varphi^{(0)})$ which solves the field equations derived from this action.

Scalar Field Equation in the Background

The equation of motion for the scalar field is:

$$\square\varphi - V'(\varphi) = 0. \quad (18)$$

If we assume a constant scalar field configuration, i.e.,

$$\varphi(x) = \varphi_0 = \text{const}, \quad (19)$$

then the kinetic term vanishes:

$$\partial_\mu \varphi = 0 \quad \Rightarrow \quad \square\varphi = 0, \quad (20)$$

and the scalar field equation reduces to:

$$V'(\varphi_0) = 0. \quad (21)$$

This condition must be satisfied by the potential for φ_0 to be an admissible constant background solution.

Einstein Equations in the Background

The Einstein field equations in the Einstein frame are:

$$\tilde{G}_{\mu\nu} = \partial_\mu \varphi \partial_\nu \varphi - \frac{1}{2} \tilde{g}_{\mu\nu} \left[(\tilde{\nabla} \varphi)^2 + 2V(\varphi) \right]. \quad (22)$$

Again, assuming $\varphi = \varphi_0 = \text{const}$ implies $\partial_\mu \varphi = 0$, so the stress-energy tensor becomes:

$$T_{\mu\nu} = -\tilde{g}_{\mu\nu} V(\varphi_0). \quad (23)$$

Therefore, the Einstein equation simplifies to:

$$\tilde{G}_{\mu\nu} = -\tilde{g}_{\mu\nu} V(\varphi_0). \quad (24)$$

This is the field equation for an Einstein space (i.e., one with constant Ricci curvature), and its solution is:

- Schwarzschild spacetime if $V(\varphi_0) = 0$,
- Schwarzschild–de Sitter or Schwarzschild–anti-de Sitter if $V(\varphi_0) \neq 0$.

Thus, taking $\varphi = \varphi_0$ constant is justified *only if* φ_0 is a *critical point* of the potential:

$$\boxed{V'(\varphi_0) = 0.} \quad (25)$$

In this case, we may adopt the standard Schwarzschild metric as the background, and proceed to study perturbations around it.

Linearized Perturbations around the Background

We now consider small perturbations around the background configuration:

$$\tilde{g}_{\mu\nu} = \tilde{g}_{\mu\nu}^{(0)} + \delta \tilde{g}_{\mu\nu}, \quad (26)$$

$$\varphi = \varphi_0 + \delta \varphi. \quad (27)$$

Since φ_0 is constant and satisfies $V'(\varphi_0) = 0$, the field equations at linear order in perturbations decouple:

Scalar Field Perturbation: Expanding the scalar field equation to linear order:

$$\square \varphi - V'(\varphi) = 0 \quad \Rightarrow \quad \delta(\square \varphi) - V''(\varphi_0) \delta \varphi = 0. \quad (28)$$

This becomes:

$$\square \delta \varphi - V''(\varphi_0) \delta \varphi = 0. \quad (29)$$

So the scalar perturbation satisfies a Klein-Gordon-type equation with an effective mass:

$$m_{\text{eff}}^2 = V''(\varphi_0). \quad (30)$$

Metric Perturbations: The metric perturbations evolve according to the standard linearized Einstein equations in vacuum (if $V(\varphi_0) = 0$), or with a cosmological constant (if $V(\varphi_0) \neq 0$).

In either case, the metric and scalar perturbations obey *decoupled equations*, and can thus be analyzed independently.

This decoupling is a key simplification that permits separate treatment of quasinormal modes (QNMs) for the scalar and tensor sectors.

Hence, both theoretically and computationally, assuming a constant background scalar field is well-justified when $V'(\varphi_0) = 0$, and it allows for a clean separation of perturbations into scalar and gravitational modes in the Einstein frame.

Quasinormal Modes of the Scalar Field in the Einstein Frame

We now consider the calculation of quasinormal modes (QNMs) for the scalar field in the Einstein frame. Having already transformed the original $f(R) = R + \alpha R^2$ gravity to the Einstein frame and identified a minimally coupled scalar field with potential $V(\varphi)$, we proceed as follows.

Background Spacetime

We assume the background metric is the Schwarzschild solution:

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad (31)$$

This assumption is justified because we previously took the scalar potential $V(\varphi_0) = 0$, which implies the scalar energy-momentum tensor vanishes at the background level. Hence, the Einstein equations reduce to vacuum Einstein equations, and Schwarzschild is a valid solution.

Perturbation of the Scalar Field

We consider small perturbations around the constant background scalar field:

$$\varphi(x^\mu) = \varphi_0 + \delta\varphi(x^\mu) \quad (32)$$

To linear order, the perturbation $\delta\varphi$ satisfies the Klein-Gordon equation in curved space-time:

$$\square\delta\varphi - m_\varphi^2 \delta\varphi = 0 \quad (33)$$

where

$$m_\varphi^2 = V''(\varphi_0) \quad (34)$$

is the effective mass squared of the scalar field in the Einstein frame.

Wave Equation in Schwarzschild Spacetime

We decompose the scalar field perturbation in spherical harmonics:

$$\delta\varphi(t, r, \theta, \phi) = \sum_{\ell m} \frac{\Psi_\ell(t, r)}{r} Y_{\ell m}(\theta, \phi) \quad (35)$$

Introducing the tortoise coordinate:

$$r_* = r + 2M \ln \left(\frac{r}{2M} - 1 \right) \quad (36)$$

the Klein-Gordon equation becomes:

$$\left(-\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r_*^2} - V_{\text{eff}}(r) \right) \Psi(t, r) = 0 \quad (37)$$

where the effective potential is:

$$V_{\text{eff}}(r) = \left(1 - \frac{2M}{r} \right) \left[\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} + m_\varphi^2 \right] \quad (38)$$

Time-Harmonic Ansatz and Schrödinger Form

We look for solutions of the form:

$$\Psi_\ell(t, r) = e^{-i\omega t} \psi_\ell(r) \quad (39)$$

This leads to the ordinary differential equation:

$$\frac{d^2\psi}{dr_*^2} + [\omega^2 - V_{\text{eff}}(r)] \psi(r) = 0 \quad (40)$$

This equation is of the Schrodinger-like form. The eigenvalues ω correspond to the quasinormal mode frequencies and are generally complex:

- $\text{Re}(\omega)$: oscillation frequency
- $\text{Im}(\omega) < 0$: damping rate

Boundary Conditions for Quasinormal Modes

To extract the quasinormal modes, we impose the following boundary conditions:

- **Ingoing wave at the horizon** ($r_* \rightarrow -\infty$):

$$\psi(r_*) \sim e^{-i\omega r_*} \quad (41)$$

- **Outgoing wave at spatial infinity** ($r_* \rightarrow +\infty$):

$$\psi(r_*) \sim e^{+i\omega r_*} \quad (42)$$

These boundary conditions lead to a discrete set of complex frequencies ω , which are the quasinormal modes of the scalar field in the Einstein frame.

Clarification on Scalon Mass in the Einstein Frame

In our earlier discussion, we considered the scalar field φ in the Einstein frame and set the potential $V(\varphi_0) = 0$ for a constant background scalar field. This naturally raises the question:

If the potential is set to zero, does it imply that the scalaron mass is also zero?

The resolution of this question requires careful interpretation of the scalar potential and how the scalaron mass arises.

Origin of the Scalar Potential in the Einstein Frame

For the specific model:

$$f(R) = R + \alpha R^2,$$

the scalar potential in the Einstein frame is obtained from the relation:

$$V(\varphi) = \frac{Rf'(R) - f(R)}{2\kappa^2 f'(R)^2}.$$

Computing this for our model:

$$\begin{aligned} f'(R) &= 1 + 2\alpha R, \\ V(\varphi) &= \frac{\alpha R^2}{2\kappa^2 (1 + 2\alpha R)^2}. \end{aligned}$$

Therefore, $V(\varphi)$ is not identically zero. It only vanishes at $R = 0$. When we say we take $V(\varphi_0) = 0$, we are specifically assuming a vacuum background with:

- Zero Ricci scalar: $R = 0$,
- A constant background field: $\varphi = \varphi_0$,
- Hence, $V(\varphi_0) = 0$.

However, the scalaron mass comes from the *second derivative* of the potential with respect to φ evaluated at φ_0 , and this need not vanish.

Computation of the Scalon Mass

To find the scalaron mass, we use the standard expression in $f(R)$ gravity:

$$m^2 = \frac{1}{3} \left(\frac{1}{f''(R)} + \frac{Rf'(R) - 2f(R)}{f'(R)^2} \right).$$

For $f(R) = R + \alpha R^2$:

$$\begin{aligned} f'(R) &= 1 + 2\alpha R, \\ f''(R) &= 2\alpha. \end{aligned}$$

Evaluating at $R = 0$:

$$\begin{aligned} f'(0) &= 1, \\ f''(0) &= 2\alpha, \\ f(0) &= 0. \end{aligned}$$

Then,

$$m^2 = \frac{1}{3} \left(\frac{1}{2\alpha} + \frac{0 - 2 \cdot 0}{1^2} \right) = \frac{1}{6\alpha}.$$

Thus, the scalaron mass is

$$m^2 = \frac{1}{6\alpha} \neq 0.$$

This confirms that the scalar field has a non-zero mass even though the potential $V(\varphi_0) = 0$ at the background. The second derivative $V''(\varphi_0)$ does not vanish and leads to a well-defined Klein-Gordon equation with a mass term.

Conclusion

It is justified and correct to say:

- The background scalar potential is zero.
- The scalar field is in stable equilibrium at φ_0 .
- The scalar field has a non-zero mass $m^2 = 1/(6\alpha)$, which controls the propagation and decay of scalar perturbations.

This confirms the validity of the Klein-Gordon equation we derived earlier, and we can now proceed to compute the QNMs for the scalaron in the Einstein frame.

Quasi-Normal Modes of the Scalaron in the Einstein Frame

We begin with the **linearized Klein-Gordon equation** for the scalar field φ in the Einstein frame background:

$$\square \delta\varphi - m^2 \delta\varphi = 0,$$

where $\square$ is the covariant d'Alembert operator with respect to the Einstein-frame metric $\tilde{g}_{\mu\nu}$, and $m^2 = \frac{1}{6\alpha}$ is the scalaron mass.

Background Geometry

We assume the background spacetime is the **Schwarzschild metric** in the Einstein frame (since we considered $R = 0 \Rightarrow \tilde{R}_{\mu\nu} = 0$):

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

Step 1: Scalar Field Ansatz

We expand the scalar perturbation in spherical harmonics:

$$\delta\varphi(t, r, \theta, \phi) = \sum_{\ell m} \frac{\psi_{\ell m}(t, r)}{r} Y_{\ell m}(\theta, \phi).$$

This decomposition separates the angular dependence and allows us to write the equation in terms of $\psi_{\ell m}(t, r)$.

Step 2: Apply the D'Alembertian Operator

Using the standard result for a scalar field in a spherically symmetric background:

$$\square\delta\varphi = - \left(1 - \frac{2M}{r}\right)^{-1} \frac{\partial^2 \delta\varphi}{\partial t^2} + \frac{\partial}{\partial r} \left[\left(1 - \frac{2M}{r}\right) \frac{\partial \delta\varphi}{\partial r} \right] + \frac{1}{r^2} \nabla_{S^2}^2 \delta\varphi,$$

where $\nabla_{S^2}^2 Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$, the angular Laplacian.

Plugging in the ansatz and simplifying, we get the equation for $\psi_{\ell m}(t, r)$:

$$-\frac{\partial^2 \psi_{\ell m}}{\partial t^2} + \left(1 - \frac{2M}{r}\right) \frac{\partial}{\partial r} \left[\left(1 - \frac{2M}{r}\right) \frac{\partial \psi_{\ell m}}{\partial r} \right] - \left(1 - \frac{2M}{r}\right) \left[\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} + m^2 \right] \psi_{\ell m} = 0.$$

Step 3: Use the Tortoise Coordinate

Define the tortoise coordinate r_* by:

$$\frac{dr_*}{dr} = \left(1 - \frac{2M}{r}\right)^{-1} \Rightarrow r_* = r + 2M \ln \left(\frac{r}{2M} - 1 \right).$$

This allows the equation to take the Schrödinger-like form. We define:

$$\psi_{\ell m}(t, r) = \Psi(r_*) e^{-i\omega t}.$$

Then, the wave equation becomes:

$$\frac{d^2 \Psi}{dr_*^2} + [\omega^2 - V_{\text{eff}}(r)] \Psi = 0,$$

where the **effective potential** is:

$$V_{\text{eff}}(r) = \left(1 - \frac{2M}{r}\right) \left[\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} + m^2 \right].$$

Final QNM Equation

The scalaron QNM equation in the Einstein frame is:

$$\frac{d^2\Psi}{dr_*^2} + \left[\omega^2 - \left(1 - \frac{2M}{r} \right) \left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3} + \frac{1}{6\alpha} \right) \right] \Psi = 0.$$

We now look for solutions $\Psi(r_*)$ that satisfy:

- **Ingoing waves at the horizon** $r_* \rightarrow -\infty$: $\Psi \sim e^{-i\omega r_*}$,
- **Outgoing waves at infinity** $r_* \rightarrow +\infty$: $\Psi \sim e^{+i\omega r_*}$.

These boundary conditions define the quasi-normal modes ω for the scalaron field.